

02741 Generative AI for Biomedicine

Assignment 2: Using Evo2 embeddings to Predict the 3D Folding of Chromatin

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1 Introduction

Predicting 3D chromatin folding from 1D DNA sequence is a challenging problem that requires learning long range dependencies across genomic regions. The DNALongBench benchmark introduced a baseline convolutional neural network (CNN) which achieved an average Pearson correlation coefficient (PCC) of 0.098 on the contact map prediction task. In this work, I evaluate whether Evo2 7B, a DNA foundation model, can yield stronger performance when paired with a lightweight prediction head trained on top of frozen Evo2 embeddings.

The assignment required testing a default model and then choosing one option from Section 3c to improve performance. I attempted two of the suggested approaches: (1) training the baseline model longer for 25 epochs and (2) replacing the linear prediction head with a multilayer perceptron (MLP) including BatchNorm and Dropout.

2 Methods

2.1 Dataset

All experiments were trained on the subset provided in the assignment: `train-0.tfr`, `valid-0.tfr`, and `test-0.tfr`. Each sample corresponds to a 40×40 chromatin contact map.

2.2 Model Setup

Evo2 7B was used in frozen mode. Only the prediction head was trained. Optimization used Adam with a base learning rate of 6×10^{-4} (except for the improved prediction head where I tried 3×10^{-4} alongside a weight decay of 10^{-4}) and a step scheduler with step size 3 and decay factor 0.1. Mean squared error was used as the loss function, and PCC averaged across the test set was used as the main evaluation metric.

2.3 Prediction Model Variants

- **Default head (10 epochs):** Single linear layer `Linear(8192, 1)`
- **Extended training (25 epochs):** Same architecture, trained longer for 25 epochs instead of 10.
- **Improved head (MLP 10 epochs):** Modified and improved prediction head architecture while maintaining 10 epochs like in the original run:

$8192 \rightarrow 2048 \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow 1.$

WandB Link

<https://api.wandb.ai/links/abanka-carnegie-mellon-university/8nevfydl>

3 Results

3.1 Quantitative Results

Model Variant	Epochs	Best Validation Loss	Avg PCC	Std Dev PCC
Linear head (default)	10	6.9496	-0.0093	0.0452
Linear head (25 epochs)	25	6.9220	-0.0092	0.0452
MLP head (BatchNorm + Dropout)	10	0.2323	0.1339	0.0528
CNN baseline (DNALongBench on HFF)	30	—	0.0980	—

Table 1: Comparison of Evo2 finetuning configurations with the DNALongBench CNN baseline.

3.2 Representative Visualizations

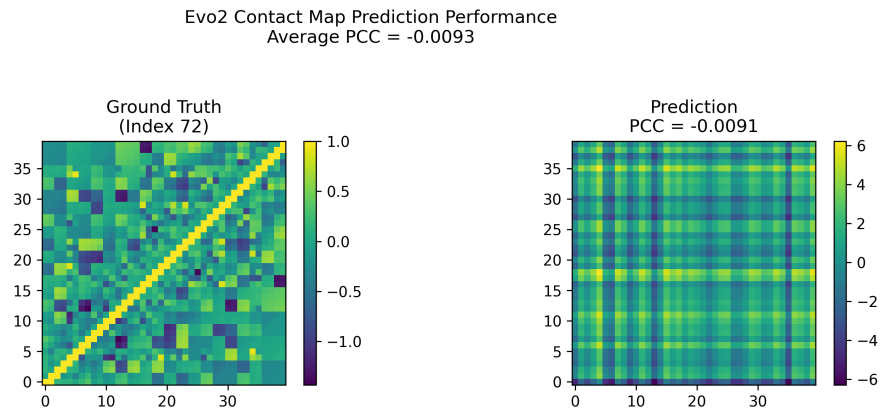


Figure 1: Representative example from the default 10 epoch linear head model. The PCC is near zero and the prediction shows no meaningful structure.

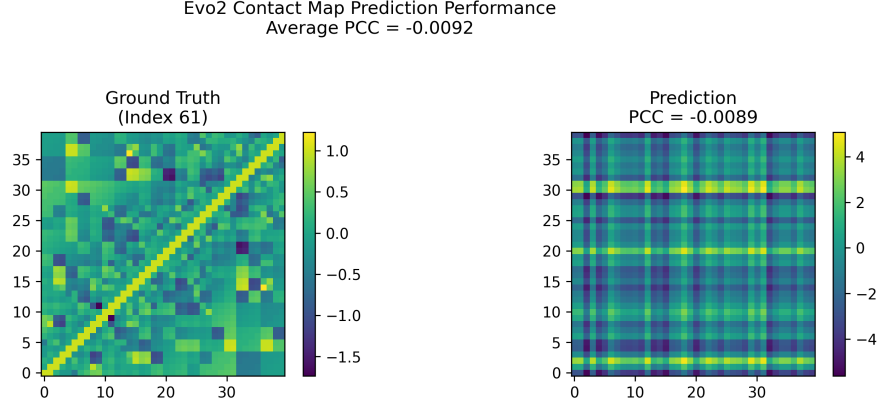


Figure 2: Representative example from the 25 epoch linear head model. Despite longer training, the model does not improve and the PCC remains close to zero.

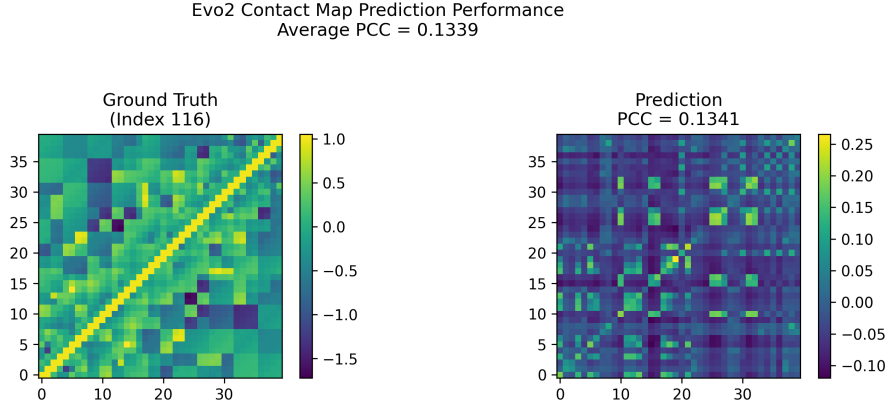


Figure 3: Representative example from the improved MLP prediction head. The predicted structure shows clearer similarity to the ground truth and achieves a PCC of 0.1341.

4 Analysis

4.1 Performance Comparison

The default Evo2 model with a linear prediction head performed significantly worse than the CNN baseline from the DNALongBench paper, achieving an average PCC of -0.0093 compared to the baseline value of 0.098 . This result indicates that the Evo2 embeddings, when passed through a single linear transformation, do not contain contact map information in a linearly separable form.

The 25 epoch version of the same model did not improve performance and remained at a PCC of approximately -0.0092 . Although the training loss continued decreasing, the validation loss increased substantially, confirming overfitting without any corresponding gain in predictive structure.

In contrast, the improved MLP head achieved an average PCC of 0.1339 , which is higher than the CNN baseline. This demonstrates that the Evo2 embeddings do encode relevant contact map structure, but this structure only becomes usable when the prediction head has sufficient nonlinearity and normalization. The MLP head improves performance by 36 percent relative to

the CNN model and is the only setting where the predicted map visually resembles the ground truth.

4.2 Discussion

The failure of the linear head and the 25 epoch run highlights a key limitation: the Evo2 embedding space is not aligned with the regression target in a purely linear way. Chromatin folding is driven by complex sequence-to-structure relationships involving enhancer-promoter looping, TAD boundaries, nucleosome positioning, and other effects that are inherently nonlinear. A linear model cannot express these dependencies, so it converges to near-random behavior. Extending training does not fix this because the model has no representational capacity to learn the mapping.

The improved MLP head partially overcomes this issue by applying feature scaling (Batch-Norm), nonlinear mixing of embedding channels (ReLU), and regularization (Dropout). This allows the model to separate useful dimension interactions in the Evo2 embedding space, explaining the jump from -0.0093 to 0.1339 PCC.

Performance is still modest relative to state of the art contact map models because (1) only a small fraction of the dataset was used, (2) Evo2 was fully frozen and never adapted to the task, and (3) the prediction head operates on flattened embedding pairs instead of learning 2D spatial structure.

Better performance could have been achieved by:

- Training on all `train-*.tfr` shards instead of a single subset.
- Using a cosine or warmup learning rate schedule instead of fixed step decay.
- Predicting only the upper triangle of the contact map to reduce redundancy.
- Replacing the MLP head with a 2D convolutional or cross-attention head.
- Unfreezing the upper Evo2 transformer blocks for task-specific finetuning.

Overall, the comparison confirms that Evo2 embeddings are useful but not directly decodable without an expressive head. The results suggest that the model is bottlenecked by the head design rather than the embedding quality.

5 Conclusion

Two improvement strategies were attempted. Increasing epochs did not help, but replacing the linear head with an MLP significantly improved PCC and exceeded the CNN baseline. This confirms that the form of the prediction head is critical when using frozen language model embeddings for structured genomics tasks.